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AI-supported Power Electronics Education

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The ECPE White Paper on ´Artificial Intelligence (AI) in Power Electronics´ was drawn up by a group of experts from the ECPE Research Network as part of an ECPE Task Force with several subgroups (SGs), which began its deliberations in September 2025. This White Paper, available on the ECPE website www.ecpe.org traces the application of AI methods and tools throughout the entire life cycle of a power electronics product – from the early development phase, through to manufacturing and finally the product´s operational phase.

The Task Force has decided to address the cross-cutting issue of the use of AI methods and tools in human learning, education and advanced training in this separate White Paper document on ´AI-supported Power Electronics Education´ prepared by subgroup SG 5.

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AI in Power Electronics Education

I. Introduction

Power electronics brings together circuit theory, switching-device physics, control, magnetics, thermal design, and embedded implementation. This makes the subject difficult to teach as a sequence of isolated topics. Students must move between averaged and switched models, analytical design equations and nonideal hardware behaviour, and simulation results and measured waveforms. In practice, teaching is limited not only by the complexity of the subject, but also by restricted laboratory time, the cost and safety constraints of converter hardware, and the difficulty of giving timely feedback on multi-step design tasks [3], [5].

In this context, AI is most useful as a support layer for teaching and assessment rather than as a substitute for subject expertise. Power electronics courses generate structured technical data that can support diagnostic feedback, adaptive learning, and assessment. The main educational value is therefore not generic text generation, but the ability to relate student work to the physical and mathematical constraints of converter operation.

Research on intelligent tutoring systems and adaptive instructional support indicates that AI-based tutoring is most effective when it models the learner's state and provides task-specific feedback rather than general hints [1], [2]. Virtual and remote laboratories provide an additional setting for AI-supported experimentation and feedback [3], [4]. The key issue is therefore where AI can provide useful instructional support without reducing physical understanding, experimental discipline, or academic integrity.

II. AI Foundations for Power Electronics Education

AI-supported power electronics education does not depend on a single technology. Intelligent simulation, adaptive assessment, automated feedback, and learning analytics draw on a small number of recurring AI method families. Establishing these foundations is important because the individual applications discussed later are not isolated tools; they rely on the same underlying methods, data structures, and domain constraints.

Power electronics has two characteristics that are particularly relevant to the use of AI. First, it is governed by well-defined physical laws. A converter either satisfies the applicable circuit relationships and operating constraints or it does not. An AI system that reasons about power-electronic systems must therefore respect these constraints rather than approximate them only from statistical patterns. Second, the field is both multimodal and multiscale. It combines schematics, waveforms, characteristic curves, layouts, and controller code with dynamics ranging from nanosecond switching events to much slower control and thermal processes. A useful educational AI system must therefore combine general-purpose AI methods with explicit domain structure.

Most educational applications can be traced to four main families of AI methods. These families differ in maturity, cost, interpretability, and data requirements. In many cases, several of them must be combined within the same application. Distinguishing between them is useful because recent attention to large language models can otherwise obscure the continued importance of more established methods. Statistical learning, for example, remains one of the most transparent and economical approaches and continues to support many learning-analytics systems, risk indicators, and early-warning dashboards. Physics-aware methods, by contrast, connect AI more directly to the engineering content and help ensure that model outputs remain consistent with the underlying system behaviour.

The four method families and their roles in power electronics education are as follows:

- *Statistical and machine-learning models:* Regression, classification, and clustering methods learn from structured data to predict performance, group recurring errors, identify learner profiles, and flag risks. These methods are generally mature, computationally inexpensive, and comparatively interpretable.
- *Large language and multimodal models:* Language models can generate explanations, review derivations, comment on reports, and analyse controller code. Multimodal variants extend these capabilities to schematics, waveforms, and other visual representations, which is important in a discipline where much of the reasoning is expressed graphically rather than only through text.
- *Physics-informed and surrogate models:* Physics-informed neural networks and related hybrid methods incorporate governing equations or physical constraints into the learning process. Data-driven surrogate models can approximate converter behaviour at substantially lower computational cost than detailed simulation. Both approaches are relevant to intelligent simulation because they can support interactive or faster-than-real-time exploration of converter operation.
- *Knowledge representation and retrieval:* Curriculum knowledge graphs can encode relationships between converter topologies, components, control methods, operating modes, and learning competences. When combined with retrieval-augmented generation, these structures can ground language-model outputs in verified sources and make the resulting explanations more traceable.

General-purpose models are often the starting point, but adapting them to power electronics education requires three additional elements. The first is grounding in verified sources, including textbooks, standards, validated reference solutions, and instructor-approved course material. This provides an authoritative basis against which generated explanations and calculations can be checked. The second is multimodal capability, so that schematics, measured waveforms, and simulation results are treated as primary inputs rather than as secondary illustrations. The third is the explicit enforcement of physical constraints, so that generated designs, explanations, and recommendations remain consistent with circuit theory and admissible operating conditions.

Annotated educational data in power electronics are limited and often protected by copyright or institutional restrictions. For this reason, domain adaptation will usually rely on modest fine-tuning combined with retrieval from a curated knowledge base rather than on large-scale model training. This approach also reduces computational cost and limits the data-protection burden for individual universities or training providers.

These foundations can be viewed as a layered stack, as illustrated in Fig. 1. The lower layer contains the general AI method families. Above it, a domain-adaptation layer provides verified sources, multimodal handling, physical constraints, and curriculum structure. Educational functions such as tutoring, assessment, simulation, experimentation, and learning analytics are then built on top of this common base.

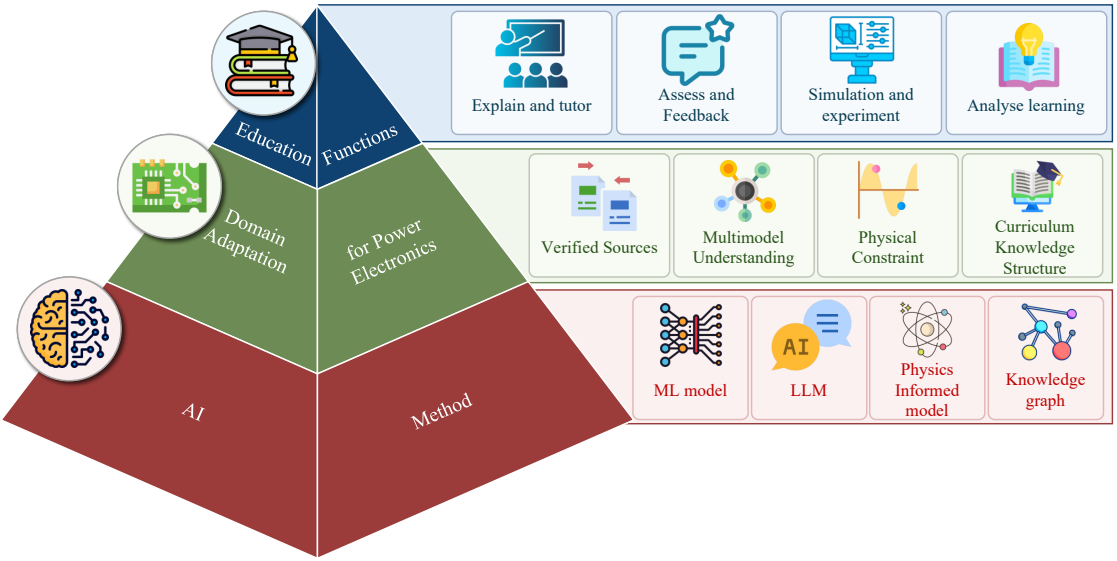


Fig. 1: Layered view of the AI foundations for power electronics education. Educational functions build on a domain-adaptation layer, which in turn builds on general AI method families.

Because many statements in power electronics can be verified against equations, operating limits, or simulation results, technical accuracy can be engineered into the system rather than assumed. A characteristic failure mode of a general-purpose model is a response that is fluent and confident but technically incorrect. In an engineering context, this is not a minor presentation issue, since a learner may not be able to distinguish a plausible explanation from a valid one.

A practical safeguard is a generate–then–verify workflow. The AI model first proposes an explanation, derivation, or design, after which an independent verification stage checks the result before it is presented to the learner. Depending on the task, this check may involve symbolic equations, circuit constraints, numerical reference calculations, simulation results, or instructor-defined rules. Such runtime verification should be complemented by systematic model evaluation. Shared benchmarks and calibrated item banks for power-electronics reasoning would make it possible to compare different systems on common technical tasks, which is currently difficult.

III. AI-Assisted Teaching and Learning in Power Electronics

Building on the foundations described in Section II, AI-assisted teaching can operate directly on student-generated technical artefacts. A useful implementation pattern is to treat the AI layer as a diagnostic engine operating on student-generated artefacts. In a converter analysis task, for example, the system can compare submitted state-space models, switching-function derivations, or simulation waveforms against a reference solution family rather than a single numerical answer. This makes it possible to identify whether an error originates from polarity convention, duty-ratio definition, device nonideality, sampling interpretation, controller discretization, or misunderstanding of continuous-conduction versus discontinuous-conduction operation. That kind of error localization is pedagogically more valuable than answer verification alone.

A second high-value use case is adaptive scaffolding in design-oriented exercises. Power electronics tasks are usually hierarchical: topology selection precedes component sizing; component sizing precedes control tuning; control tuning precedes

transient and thermal verification. AI support can exploit this dependency structure to release hints selectively. Instead of telling the student how to solve the full problem, the system can detect which prerequisite subproblem has failed and redirect attention to that stage. In a buck-converter design exercise, for instance, the tutor can infer from ripple current, duty-cycle range, or loop bandwidth errors whether the student is struggling with inductance sizing, switching-frequency tradeoffs, or small-signal control assumptions. This is close to the instructional logic of intelligent tutoring systems, but here the domain model must include power-electronic constraints and admissible operating regions [1], [2]. The interaction between these elements can be viewed as a closed instructional loop in which student work is interpreted against a domain model, used to update the learner model, and translated into an appropriate instructional response, as illustrated in Fig. 2.

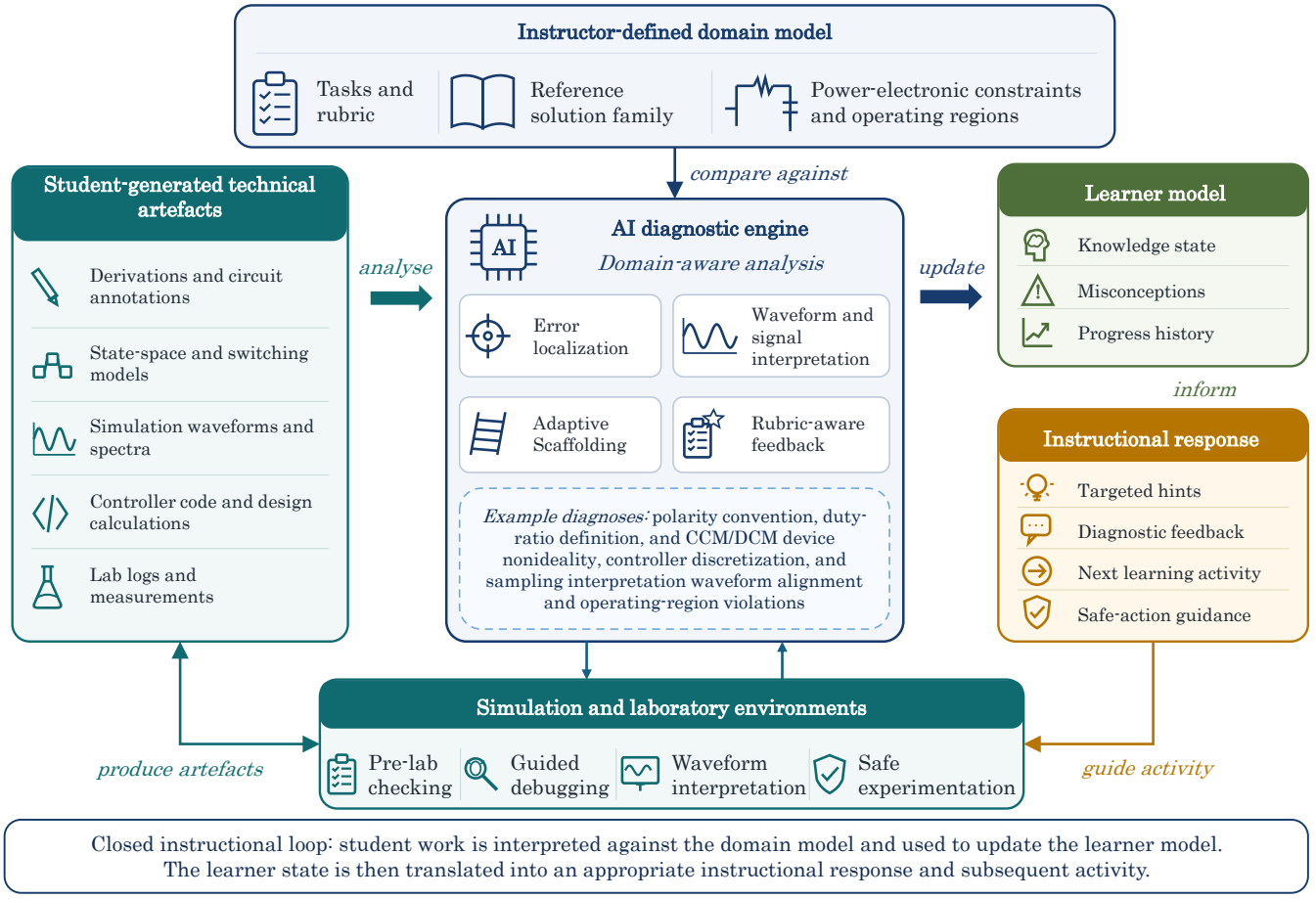


Fig. 2: Conceptual framework for AI-assisted teaching and learning in power electronics. Student-generated technical artefacts are analysed against an instructor-defined domain model and used to support diagnostic feedback, adaptive scaffolding, and selection of subsequent learning activities [9].

Simulation and laboratory environments provide a further source of structured technical data for AI-supported diagnosis and feedback. Their use for pre-lab checking, waveform interpretation, guided debugging, and safe experimentation is discussed in Section IV.

Generative AI is also relevant, but only in narrowly constrained roles. In power electronics, unrestricted text generation is far less useful than domain-bounded generation. The more defensible use cases are the generation of parameterized problem variants, first-draft laboratory instructions, rubric-aligned formative comments, and guided explanations linked to equations or waveforms already present in the task. Used this way, generative models can reduce authoring effort while remaining subject to the grounding and verification principles described in Section II.

To keep the section compact, Table I summarizes implementation roles that are technically useful in power electronics education without repeating the discussion above.

IV. Intelligent Simulation and Virtual Laboratories

Simulation and laboratory work are central to power electronics education because switching transients, thermal limits, control interactions, and nonideal device behaviour are difficult to understand from analytical equations alone. Physical laboratories represent these effects directly, but their use is limited by equipment cost, safety requirements, laboratory capacity, and the

TABLE I: AI roles in power electronics education and the corresponding technical input artefacts

AI role	Primary artefact	Output type	Main technical constraint
Concept-diagnostic tutor	Derivations, circuit annotations, symbolic models	Error localization, targeted hints	Domain rule encoding
Waveform feedback engine	Time-domain traces, spectra, switching states	Misconception classification	Signal alignment and labeling
Adaptive exercise sequencer	Quiz history, lab logs, simulation outcomes	Difficulty progression	Reliable learner-state estimation
Lab safety assistant	Command sequence, setpoints, interlock states	Unsafe-action blocking	Hard constraint enforcement
Rubric-aware feedback generator	Reports, code, design calculations	Formative comments	Grounding to instructor rubric

risk of damaging hardware. Conventional circuit simulation removes some of these constraints, although the simulator itself remains passive: it computes the model specified by the learner but does not explain why a design fails or which aspect of the result should be examined. AI can extend simulation in two related directions. First, it can reduce the computational effort required for repeated or interactive studies. Second, it can add an instructional layer that interprets the learner's actions and provides guidance during the experiment. The result is not simply a faster simulator, but an environment in which parameter changes, operating limits, and design trade-offs can be explored with structured feedback. Three developments are particularly relevant.

- *Reduced-order and surrogate models:* Machine-learning models can approximate converter, thermal, or system-level behaviour at a fraction of the computational cost of detailed switched simulation. This makes interactive exploration possible, allowing learners to vary a parameter and observe its effect without waiting for a complete simulation run. Such models are especially useful for repeated sweeps, sensitivity studies, and real-time visualisation, provided that their valid operating range is clearly defined.
- *Digital twins:* A calibrated digital twin provides a virtual representation of a converter or power-electronic system using models that are updated or validated against measured data. Learners can vary load, switching frequency, control parameters, and environmental conditions, inject disturbances, and examine the corresponding electrical and thermal response. This creates a useful link between idealised simulation and the behaviour of physical hardware.
- *Guided design-space exploration:* AI can support the selection of parameters and operating points to be investigated. Rather than leaving learners to search a large design space without direction, the system can identify unstable regions, constraint boundaries, and operating points that illustrate relevant trade-offs, such as efficiency versus switching frequency or transient response versus control bandwidth.

Figure 3 shows a possible architecture. The learner interacts with an AI-supported instructional layer, while the underlying numerical work remains within the simulation, surrogate-model, digital-twin, or hardware-in-the-loop environment. The AI layer can generate scenarios, interpret intermediate results, and select suitable hints, but it should not replace explicit solver checks, safety limits, or validated reference models.

A virtual laboratory extends this concept by managing the experimental process rather than only accelerating the simulation. It can define the task, monitor the sequence of learner actions, request predictions before measurements are taken, and intervene when the procedure becomes unsafe or technically unproductive. In this role, the system performs some of the routine guidance normally provided by a laboratory demonstrator, while leaving technical responsibility and final evaluation with the instructor. The most useful capabilities are the following:

- *Scenario generation and fault injection:* The system can create parameterised operating conditions and introduce realistic faults or disturbances, such as excessive dead time, unstable controller settings, sensor offsets, load steps, or thermal-limit violations. Learners can then diagnose the cause from the resulting measurements.
- *Procedural scaffolding:* The AI tutor can guide the experimental sequence, request hypotheses, suggest measurements, and provide progressively more explicit hints. Safety-related interventions should be immediate, whereas conceptual guidance should allow the learner time to inspect the evidence and revise the approach.
- *Integration with engineering tools:* The laboratory should connect to established software and hardware environments such as LTspice, PLECS, MATLAB/Simulink, digital control platforms, and hardware-in-the-loop systems. This keeps the learning task close to the workflows used in industrial practice and avoids isolating the AI component from the engineering toolchain.

These environments are particularly useful when access to physical equipment is limited. They allow learners to practise measurement planning, controller adjustment, fault diagnosis, and interpretation of converter behaviour before working on hardware. They also provide a controlled setting in which serious design or operating errors can be examined without risk to equipment. Their educational value, however, depends on preserving the distinction between simulation and experiment: a virtual laboratory should prepare learners for physical measurements rather than create the impression that a model fully represents real hardware.

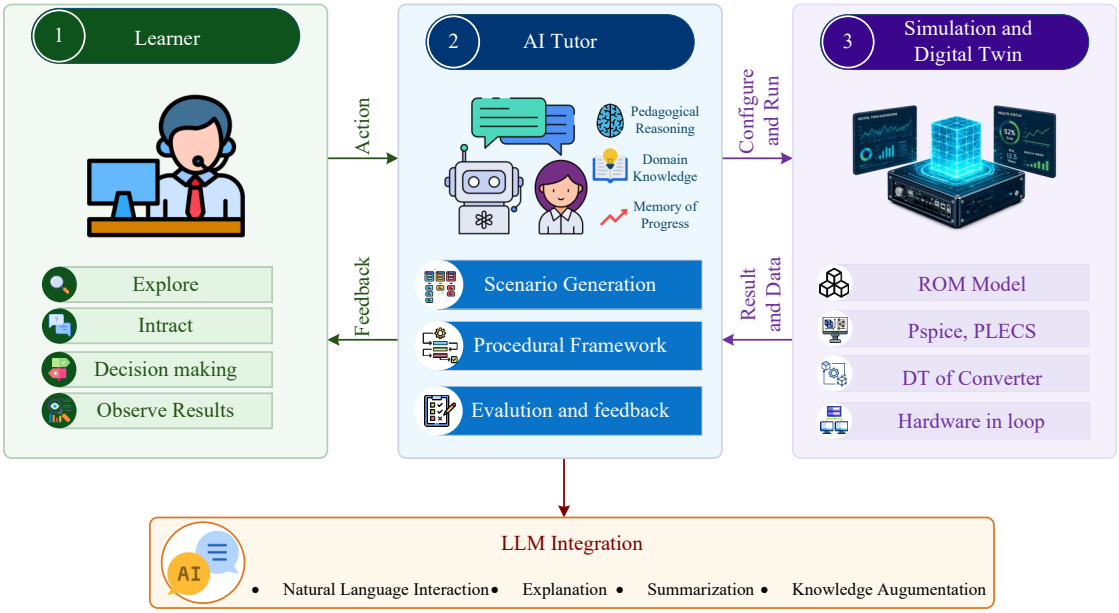


Fig. 3: Architecture of an AI-supported virtual laboratory for power electronics education. The instructional layer mediates between the learner and the simulation, digital-twin, or hardware-in-the-loop environment by generating scenarios, monitoring progress, and providing grounded feedback.

V. Adaptive Assessment and Automated Feedback Systems

A. Context and Motivation

Learners enter power electronics courses with heterogeneous prior knowledge, both in undergraduate programmes and in industrial continuing education. Traditional one-size-fits-all assessment and exercise formats serve this heterogeneity poorly: they under- or over-challenge sub-groups, usually provide feedback late and summatively, and consume substantial teaching capacity for grading. Adaptive assessment and AI-supported feedback address exactly these weaknesses. Adaptive assessment and AI-supported feedback can provide data-driven individualisation, timely formative feedback, and reduced repetitive grading and authoring workload.

B. Adaptive Assessment and Grading Methods

Adaptive assessment (Computerized Adaptive Testing, CAT) selects the next item based on previous responses. It is often grounded in Item Response Theory (IRT): items are calibrated by parameters such as difficulty, discrimination and guessing probability, and the learner's ability is estimated continuously. This yields a more accurate and fairer competence estimate with fewer items, because items are targeted at the individually most informative difficulty range.

For power electronics, the following approaches are particularly viable:

- *Competence- and outcome-oriented grading:* items are tagged along a learning-outcome taxonomy (e.g. Bloom: understand, apply, analyse, evaluate, design) and mapped to a competence model, producing not just a score but a differentiated competence profile (e.g. "output filter design: solid" vs. "switch loss estimation: incomplete").
- *Knowledge tracing:* models such as Bayesian Knowledge Tracing or more recent learning-curve-based methods estimate mastery of individual sub-competences over time and drive item selection accordingly.
- *Parametric item generators:* from one item template (e.g. buck-converter design), varying the parameters – input voltage, switching frequency, ripple targets, load range – produces virtually unlimited, individually distinct variants with a uniquely computable reference solution. This discourages illegitimate copying while enabling targeted practice of specific competences.

Careful calibration of the item pool is essential: item parameters must be estimated on a sufficient data basis and regularly checked for drift. Especially in small cohorts (typical of advanced power-electronics modules), hybrid approaches that combine expert estimates from teachers with the available response data are advisable.

C. AI-Generated Automated Feedback

The most immediate benefit of generative AI in teaching lies in formative, in-process feedback. Whereas classical e-assessment systems could only auto-grade closed formats (multiple choice, numeric entry), large language models for the

first time enable robust, explanatory feedback on open answers, derivations and design decisions – instantly, around the clock, and in the learner's own language.

Typical use cases in power electronics:

- *Grading derivations and solution paths:* the system checks not only the final result but identifies the step at which an error occurs (e.g. a wrongly applied volt-second balance in inductor sizing) and gives a targeted hint rather than the finished solution.
- *Feedback on simulation results:* key figures can be extracted from the output of common tools (e.g. LTspice, PLECS, GeckoCIRCUITS) and compared against target values; the system comments on anomalies (excessive switching losses, unstable control, wrong dead-time) and links them to the underlying principles.
- *Grading of lab reports and designs:* rubric-based scoring against predefined criteria (completeness, plausibility, error discussion), including linguistic and structural hints.
- *Review of control or embedded code* (e.g. C or HDL for digital controllers): hints on typical errors such as wrong scaling, saturation, or sampling-time issues.

Automated feedback should follow the grounding and verification principles introduced in Section II. A clear separation between formative (learning-supportive) feedback and summative (grade-relevant) assessment is mandatory – the latter remains under human responsibility.

D. Personalisation of the Learning Process

Building on the competence estimate and the feedback, individual learning paths can be designed. An adaptive system diagnoses prior-knowledge gaps (e.g. missing fundamentals in Fourier analysis or control theory), proposes targeted preparatory units, and adapts sequence, difficulty and example context. Intelligent tutoring systems can guide learners through circuit design in dialogue without giving away the solution – comparable to an always-available tutor.

This personalisation is especially valuable for the ECPE tutorial for the continuing education of industry engineers envisaged by the Task Force: participants with very different backgrounds can take the same course at an individually appropriate depth and pace. Power-electronics competence thus becomes scalable to teach – a direct contribution to the TF objective of strengthening the education and training of PE professionals.

E. Efficiency for Educators

Adaptive assessment and automated feedback are a major lever not only for learners but also for educators. Concrete relief effects:

- *Item authoring:* AI-supported generation and variation of exercise and exam items including reference solutions and grading rubrics; teachers curate and verify rather than create everything themselves.
- *Grading:* pre-grading of open answers with justification; the teacher confirms, corrects or overrides (human-in-the-loop). Effort shifts from routine grading to quality assurance.
- *Teaching/learning analytics:* dashboards reveal, in aggregate, which concepts a cohort struggles with (e.g. commutation, loss calculation), enabling targeted adjustment in class and early identification of at-risk learners.

The prerequisite for genuine efficiency gains is a well-designed human-in-the-loop process and continuous quality control (sampling, calibration, appeal procedures). Otherwise the workload merely shifts from grading to checking automated outputs, without real relief.

F. Reference Architecture and Practical Implementation

For a robust implementation, a modular architecture that integrates into existing university or continuing-education IT is recommended:

- Integration into the learning management system (e.g. Moodle, ILIAS) via established standards such as LTI for tool integration and xAPI/Caliper for learning-process data.
- Clear data flows: item pool and calibration data, learner model, feedback service (language model with stored reference solutions) and analytics layer as separate, auditable components.
- Data minimisation and data sovereignty from the outset: pseudonymised learner profiles, clear deletion concepts, preferably locally or European-hosted models (see Section VI).

Such an architecture allows a gradual start – e.g. formative feedback in a single module first – and later scaling without locking in to a single vendor.

VI. Challenges, Ethics, and Implementation of AI in Power Electronics Education

A. Technical and Pedagogical Challenges

As discussed in Section II, power electronics combines multimodal technical artefacts with explicit physical constraints. The implementation challenge is therefore whether AI can interpret these artefacts reliably and support, rather than simplify, the formation of engineering competence.

A second challenge is model reliability under domain-specific ambiguity. In power electronics, multiple solution paths may be valid depending on assumptions about parasitics, switching frequency, modulation strategy, or acceptable ripple and thermal margins. That means AI-assisted feedback must be constraint-aware rather than template-matching. Otherwise, the system will penalize technically valid alternative solutions or, worse, validate incorrect ones that resemble the training examples. This issue becomes more serious in open-ended design tasks, where the distinction between under-designed, over-designed, and optimally designed solutions is contextual and often multiobjective.

Using AI in power electronics education imposes specific technical requirements beyond generic educational applications:

- *Data scarcity and domain adaptation:* high-quality, annotated teaching and assessment data for power electronics are scarce and often copyright-bound. Models must be domain-adapted and evaluated; jointly created, shared item and grading datasets (within ECPE) would add considerable value.
- *Integration and operation:* connecting to heterogeneous university IT, model versioning, reproducibility of grading and long-term availability are non-trivial operational topics.

B. Academic Integrity and Assessment Design

The ethics discussion is therefore not limited to plagiarism. Academic integrity is certainly central, especially with generative AI systems capable of producing polished reports and code, but in power electronics the more subtle risk is epistemic outsourcing: students may submit technically plausible reasoning that they cannot reproduce, defend, or adapt under changed operating conditions. Recent higher-education literature consistently identifies academic integrity, transparency, privacy, bias, and overreliance as the dominant concerns in generative-AI deployment [6]–[8]. In a laboratory-heavy discipline, these concerns should be addressed through assessment design rather than detection alone. The most robust approach is to require traceability between intermediate artefacts and final claims: for example, design justification linked to waveform evidence, controller tuning linked to measured transient response, and oral defense linked to submitted simulation files.

Plagiarism and academic integrity: generative AI undermines classical "home assignments" as an assessment format. Reliable AI detectors do not exist; bans alone are unenforceable. A more effective response is to redesign assessment towards competence- and process-oriented formats (oral exams, defence of designs, lab-bound and individualised tasks), where the learner's own demonstrated competence is central and the competent use of AI tools can explicitly be part of the learning outcome.

Finally, implementation succeeds only if the instructional contract is explicit. Students need to know what forms of AI assistance are permitted, which artefacts must remain demonstrably their own, and how technical understanding will be verified beyond generated output. In power electronics education, that usually means keeping some combination of viva-style questioning, live debugging, lab performance checks, and handwritten or controlled-environment derivations. The objective is not to exclude AI, but to ensure that AI-supported work remains coupled to demonstrable engineering judgment.

C. Data Protection and Legal Framework

Data governance also deserves attention. AI platforms used in technical courses often process student code, circuit files, waveform images, and laboratory records. These are not merely educational records; they may expose problem-solving habits, mistakes, and in some cases proprietary toolchains or institutional datasets. Any deployment should therefore define where the data are stored, whether prompts and uploads are retained for model training, how long artefacts remain accessible, and whether students have a non-AI pathway for completing assessed work. Privacy and governance are not administrative add-ons here; they shape the admissible architecture of the system.

Learning data are personal and, in part, especially sensitive data. In Europe, the General Data Protection Regulation (GDPR) applies, with principles such as data minimisation, purpose limitation and transparency. In addition, the EU AI Act classifies AI systems used to assess learning outcomes and to steer access to education as high-risk applications, imposing heightened requirements on risk management, documentation, human oversight and transparency.

Practical consequences for universities and training providers:

- Preference for locally or European-hosted models or on-premise solutions to avoid data transfer to third countries.
- Data protection impact assessment before productive use, clear consent, retention and deletion concepts.
- Right to human decision: grade-relevant decisions must not be made by automated means alone.

D. Ethical Aspects

Bias and fairness: AI systems can reproduce or amplify existing biases. In grading, this may disadvantage certain groups – for example through language bias against non-native speakers or through unbalanced training data. Fairness checks, the ability to appeal assessments, and deliberately diverse item design are required.

Copyright and intellectual property: three levels must be distinguished – the rights to the training and input data (teaching materials, lecture notes, third-party exam items), the legal status of generated content, and the licence terms of commercial models. Third-party teaching material must not be fed into AI systems unchecked; for own and jointly created content, clear (ideally open) licences should be agreed.

Transparency, accountability and deskilling: AI outputs must be identifiable as such; technical and legal responsibility remains with the teachers and the institution. A distinct risk is over-reliance – neither learners nor teachers should delegate core competences to the machine. AI should build understanding, not replace it.

E. Administrative and Organisational Challenges

In practice, adoption rarely fails on technology alone, but on organisational factors:

- *Procurement and cost:* licence, operating and compute costs, vendor lock-in, evaluation of open-source alternatives.
- *Governance:* anchoring in examination regulations and institutional policies, involvement of data-protection officers and staff representatives, clarification of liability.
- *Educator qualification:* AI literacy is a prerequisite. Without accompanying training and incentives, acceptance stays low.
- *Change management:* gradual introduction, transparent communication with learners, continuous evaluation.
- *Sustainability:* the energy and resource footprint of large models must be weighed; often smaller, specialised models suffice.

F. Implementation Roadmap

A staged, low-risk approach is recommended:

1) *Phase 0 – Governance and preparation:* guidelines, data protection impact assessment, selection of a clearly delimited pilot module, definition of success criteria.
2) *Phase 1 – Low-risk pilot:* use exclusively for formative, non-grade-relevant feedback in one module; close supervision and evaluation.
3) *Phase 2 – Scaling and integration:* connection to the learning management system, build-up and calibration of the item pool, educator training, extension to further modules.
4) *Phase 3 – Supervised adaptive summative elements:* controlled use of adaptive assessment elements with ultimate human responsibility and continuous fairness and quality checks.

A maturity model that describes the status per module/institution (from "AI-free" via "formative AI feedback" to "supervised adaptive assessment") is helpful, enabling comparability and realistic target-setting.

G. Recommendations for ECPE and the Industry–Academia Network

From the perspective of Sub-Group 5, there are concrete links to the objectives of the Task Force:

- *Shared resources:* building a calibrated, licence-clear item and grading pool for power electronics within the ECPE network – an effective lever against data scarcity.
- *Benchmark and standardisation:* common evaluation criteria for AI-supported feedback in power electronics, to make technical reliability comparable.
- *Tutorial and workshops:* use the planned ECPE tutorial as a pilot for adaptive, personalised continuing education; AI-supported teaching as a topic for one of the ECPE workshops planned for 2026.
- *Research funding:* position AI-supported engineering education as a research direction in public funding programmes – in line with the TF objective of guiding research directions.

VII. Conclusion

Artificial intelligence has growing relevance in power electronics education, particularly where teaching and learning involve the physical and mathematical constraints that govern converter operation. Power electronics combines equations, circuit models, control structures, waveforms, measurements, and phenomena occurring across different time and system scales. Effective AI-assisted learning therefore depends on technically verified sources, appropriate interpretation of multimodal engineering information, and clear incorporation of physical and design constraints. The generate-then-verify workflow presented in Section II provides a practical framework for this process by linking AI-generated outputs with engineering verification and technical reasoning.

Across teaching, simulation, and assessment, the most relevant applications include identification of errors in analytical

derivations, guidance during hierarchical converter design, automated interpretation of simulation and experimental results, and accelerated analysis using surrogate models and digital twins. These applications can support more individualized learning while maintaining technical evaluation as an integral part of the educational process. Their wider adoption will depend on the availability of suitable educational datasets, integration with institutional IT infrastructure, compliance with GDPR requirements, and implementation within the regulatory framework established for educational AI under the EU AI Act.

For ECPE and its wider network, the immediate opportunities include the development of shared and licence-clear item and grading pools, common benchmarks for evaluating AI-supported feedback, and the use of the planned tutorial as an initial platform for adaptive continuing education. These activities can provide a practical basis for evaluating AI methods in realistic power electronics training environments and for establishing common practices across the community.

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